# Agodi David Uchenna

**Frontend Engineer · React · React Native · AI Interfaces**

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## Professional Summary

Frontend Engineer who ships production React, Next.js, TypeScript and React Native interfaces, most recently the client for Nivram, a live AI assistant used by a large user base. Built the streaming pipeline that turns partial model output into 70+ reviewable action cards while holding 60 fps. Mechatronics background in embedded systems and control, so comfortable where physical systems meet the web.

## Skills

**Frontend:** React, Next.js, TypeScript, JavaScript (ES6+), React Native, Expo, Cross-platform (iOS, Android), HTML5, CSS3, Tailwind CSS, Framer Motion, Responsive Design, Design systems / design tokens
**AI interfaces and realtime:** LLM-powered chat interfaces, Streaming UI, WebSockets, Socket.IO
**State and data:** Redux Toolkit, React Query, Context API, React Hook Form, REST APIs
**Testing and tooling:** Playwright, Jest, Git, GitHub, Vercel, CI/CD, Code review, Performance optimisation, Figma, Agile / Scrum
**Backend and cloud:** Node.js, Python, Flask, Firebase (Firestore, Cloud Functions, Authentication), Paystack, Webhooks
**Embedded and graphics:** Raspberry Pi, Arduino, OpenCV, Control systems, Sensor integration, WebGL, GLSL, 3D design and printing

## Work Experience

**Frontend Engineer**, BrandDrive, Nigeria (hybrid) | 11/2025 – 08/2026
*Finance and business-data platform for growing businesses*

- Cut re-renders by ~90% at peak load (600 tokens per second collapsed to 60 renders) and held a steady 60 fps, by coalescing streamed tokens into one render per animation frame in the Socket.IO pipeline behind Nivram, the company's live AI assistant.
- Shipped 70+ reviewable action cards that turn partial model output into actions calling the same production endpoints as the app's own screens, with a PIN gate that merges authorisation into the payload before a card mounts.
- Unified theming across a 12-person React Native codebase by retrofitting one token system onto 100% of existing screens and removing per-screen colour constants.
- Delivered invoicing, an app-lock overlay with biometric and passcode fallbacks, and Android keyboard and inset handling that measures device geometry instead of assuming it.
- Verified features with Playwright (Chromium) end-to-end tests covering happy paths, failure paths and responsive layouts, plus Jest unit tests.

**Web Development Intern**, Zidio Web Development, Remote | 07/2024 – 10/2024

- Delivered responsive interfaces in HTML, CSS and JavaScript, with every change reviewed through pull requests in a shared Git repository.

**STEM Collaborator**, 3D Robotics, Nigeria | 06/2024 – 10/2024

- Integrated web technologies with cloud platforms to build interactive STEM learning tools.

**Frontend Developer Intern**, Juray Smart Solutions Ltd, Nigeria | 02/2024 – 06/2024

- Built and maintained client-facing React interfaces against REST APIs, turning Figma handoffs into responsive components.

**Robotics Intern**, Baun Robotics Educational Solutions, Nigeria | 06/2022 – 10/2022

- Programmed and assembled robotics teaching kits and ran hands-on STEM workshops for secondary-school students.

## Selected Projects

**Nivram, AI Assistant Client** | React, React Native, Socket.IO

- Chose Socket.IO streaming after React Native's fetch proved to expose no readable response body, ruling out standard web streaming on one target platform.
- Eliminated half-parsed-payload bugs by rendering pressable controls only from a complete object while still streaming prose as markdown, and hardened chunk merging for both delta and cumulative payloads so an in-flight reply survives a draft chat becoming a server thread.

**BrandDrive Mobile** | React Native, Expo, live fintech app | apps.apple.com/app/branddrive/id1623910229

- Rebuilt the transfers interface and its internal logic, and shipped credit notes and the personal home screen.
- Designed the personal dashboard around its three empty states (no wallet, no transaction PIN, negative balance), the ones a new user meets first.

**WhitePenguin, Multi-Tenant Preschool Platform** | React, Firebase | live: app.whitepenguin.co

- Removed client-controlled tenant routing, and the cross-tenant data bugs that come with it, by resolving each school from the subdomain host.
- Secured shared-tablet kiosk mode behind two identity layers (4-digit PIN or scanned code per person), with a logout that flushes pending attendance before clearing session keys.

**Sanctuary Altar of Christ Ministry** | React, TypeScript, Firebase, Paystack | live: sanctuaryaltar.org

- Led development as sole engineer across the public site, admin dashboard and donation flow (116 commits, 12 managed collections), live and maintained for 8 months.
- Closed a payment-recording gap found in an audit of my own flow, where a browser closing mid-redirect could take money without recording it, by moving payment authority to a signature-verified Cloud Function webhook.
- Collapsed retried webhook deliveries to one stored record by keying each donation on its Paystack transaction reference, and enforced authorisation in Firestore security rules instead of route guards.

**Face-Recognition Door Access (final-year project)** | React, Python, Flask, Raspberry Pi

- Achieved 1.3 s face matches and 800 ms end-to-end remote unlock by running recognition entirely on-device on a Raspberry Pi 4, so no face data leaves the device (privacy by design).
- Prevented a dropped request from leaving a door open by running auto-lock on a background deadline in a Flask API that drives a solenoid lock through a relay.
- Deployed a React dashboard on Vercel with a live MJPEG feed, remote unlock, face enrolment and an access log.

## Education

**Master of Engineering (M.Eng.), Mechatronics, Robotics, and Automation Engineering**, Vilnius Gediminas Technical University (Vilnius Tech), Lithuania | 09/2026 – Present

**Bachelor of Engineering (B.Eng.), Mechatronics Engineering**, Afe Babalola University, Ado-Ekiti, Nigeria | 2020 – 2025
Coursework: Embedded Systems, Control Systems, Instrumentation, Heat Transfer

## Languages

English: mastery (CEFR C2)